**1.** In general, given a normal vector $\mathbf{a}$ for a hyperplane, $\mathbf{a}\cdot(\mathbf{x}-\mathbf{x}_0)=0$ for any two points $\mathbf{x}$ and $\mathbf{x}_0$ on the plane (since $\mathbf{x}-\mathbf{x}_0$ is a vector on the plane and hence perpendicular to $\mathbf{a}$). So $\mathbf{a}\cdot\mathbf{x}=\mathbf{a}\cdot\mathbf{x}_0$. If we let $\mathbf{x}=(x_1,x_2,x_3)$ then $\mathbf{a}\cdot\mathbf{x}=\mathbf{a}\cdot\mathbf{x}_0$ gives a general solution when we know a single point on the plane $\mathbf{x}_0$.

**a.** The direction vector is $(3,2)$. So a normal vector $(-2,3)$ and the point $(-1,2)$ gives the equation

$$\begin{aligned}(-2,3)\cdot\mathbf{x} &= (-2,3)\cdot(-1,2)=8\\ \Rightarrow(-2,3)\cdot(x_1,x_2,x_3) &= \underline{-2x_1+3x_2=8}\end{aligned}$$

**c.** A normal vector to the plane is $(1,2,-2)$ (given by the direction vector of the orthogonal line). Using the given point $(2,0,1)$ we have,

$$\begin{aligned}(1,2,-2)\cdot\mathbf{x} &= (1,2,-2)\cdot(2,0,1)=0\\ \Rightarrow(1,2,-2)\cdot(x_1,x_2,x_3) &= \underline{x_1+2x_2-2x_3=0}\end{aligned}$$

**d.** A normal vector will be will be orthogonal to both $(1,1,1)$ and $(2,1,0)$. So then $\mathbf{a}\cdot(1,1,1)=0\Rightarrow a_1+a_2+a_3=0$ and $\mathbf{a}\cdot(2,1,0)=0\Rightarrow 2a_1+a_2=0$. Substituting $a_2=-2a_1$ into the first equation we get $a_3=a_1$. So then $\mathbf{a}=(1,-2,1)$ works. A Cartesian equation is then

$$(1,-2,1)\cdot\mathbf{x}=(1,-2,1)\cdot(1,1,2)=1\Rightarrow\underline{x_1-2x_2+x_3=1}$$

**3.** **a.** $x_1-2x_2+3x_3=4\Rightarrow x_1=2x_2-3x_3+4$. So then

$$\begin{aligned}\mathbf{x} &= (2x_2-3x_3+4,x_2,x_3).\\ &= \underline{(4,0,0)+x_2(2,1,0)+x_3(-3,0,1)}\end{aligned}$$

**c.** $x_1+x_2-x_3+2x_4=5\Rightarrow x_1=5-x_2+x_3-2x_4$. So then

$$\begin{aligned}\mathbf{x} &= (5-x_2+x_3-2x_4,x_2,x_3,x_4)\\ &= \underline{(5,0,0,0)+x_2(-1,1,0,0)+x_3(1,0,1,0)+x_4(-2,0,0,1)}\end{aligned}$$

**5.** Alternatively to the methods shown, you could use the techniques discussed in 1.4.

**a.** We have that $x_1+x_2+x_3=1\Rightarrow x_1=1-x_2-x_3$ substituting this into the other equation we have, $2(1-x_2-x_3)+x_2+2x_3=1\Rightarrow x_2=1$. By substituting this into the first equation we see that $x_2=1\Rightarrow x_1=-x_3$. So then we have,

$$\begin{aligned}\mathbf{x} &= (x_1,1,-x_1)\\ &= \underline{(0,-1,0)+x_1(1,0,-1)}\end{aligned}$$

**b.** $x_1-x_2=1\Rightarrow x_1=1+x_2$ substituting this into the other equation we have. $(1+x_2)+x_2+2x_3=5\Rightarrow x_2=2-x_3$. By substituting this into the first equation we see that $x_1=3-x_3$. Now all the variables are expressed in terms of $x_3$ so we have,

$$\begin{aligned}\mathbf{x} &= (3-x_3,2-x_3,x_3)\\ &= \underline{(3,2,0)+x_3(-1,-1,1)}\end{aligned}$$

Alternatively you might have solved for $x_2$ and gotten $\mathbf{x}=(1,0,2)+x_2(1,1,-1)$. It is also typical to write the scalar as $t$ or $s$ instead of $x_i$ so that it would not be confused as a dependent parameter.